

PLAUEN, Germany

WMO: 105690

Lat: 50.4819N

Lon: 12.1300E

Elev: 1276

StdP: 14.03

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.1	10.0	-4.7	4.5	6.3	3.4	6.9	10.7	24.7	42.2	21.6	38.5	3.2	230	0.591

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	83.4	65.2	79.6	63.7	76.5	62.1	66.8	79.5	64.9	76.7	63.3	74.0	7.5	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.9	86.7	72.5	60.3	82.0	70.2	58.8	77.6	68.0	32.0	79.8	30.5	77.0	29.3	74.3	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.7	18.1	16.1	DB	-1.9	90.4	9.3	3.2	-8.6	92.7	-14.0	94.6	-19.2	96.4	-26.0	98.7	(4)
(5)			WB	-2.8	70.3	9.2	2.2	-9.5	71.9	-14.9	73.1	-20.1	74.4	-26.8	76.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	46.2	30.5	29.8	37.7	44.1	52.7	58.2	62.4	62.3	55.9	47.9	38.2	33.6	(6)
(7)		DBStd	13.70	9.96	9.89	7.88	6.74	6.48	6.19	5.61	5.91	5.93	7.24	6.77	7.58	(7)
(8)		HDD50	2809	605	565	383	201	49	5	0	0	12	125	356	508	(8)
(9)		HDD65	6955	1070	985	845	628	382	218	119	122	278	532	804	972	(9)
(10)		CDD50	1426	0	1	3	23	133	251	385	382	188	58	2	0	(10)
(11)		CDD65	98	0	0	0	0	1	14	39	39	4	1	0	0	(11)
(12)		CDH74	1274	0	0	0	1	39	206	496	465	64	3	0	0	(12)
(13)		CDH80	289	0	0	0	0	1	41	115	128	4	0	0	0	(13)
(14)	Wind	WSAvg	7.3	9.3	8.1	8.0	7.2	6.2	6.1	5.8	5.9	6.6	7.4	8.6	8.8	(14)
(15)	Precipitation	PrecAvg	28.9	2.0	1.8	2.0	1.7	2.5	3.0	3.5	3.0	2.6	2.2	2.3	2.3	(15)
(16)		PrecMax	37.9	3.5	4.1	4.9	3.9	5.8	5.5	6.0	8.1	4.9	5.1	5.5	4.8	(16)
(17)		PrecMin	21.3	0.1	0.4	0.5	0.2	0.4	1.1	1.0	0.8	1.0	0.7	0.0	0.8	(17)
(18)		PrecStd	4.3	0.9	0.9	1.1	0.8	1.3	1.1	1.3	1.7	0.9	1.0	1.2	1.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.8	56.9	63.9	70.6	77.8	85.0	87.9	88.9	79.7	71.4	57.4	53.1	(19)
(20)			MCWB	45.8	48.2	50.9	54.2	61.3	64.2	67.9	66.8	63.2	59.3	50.4	48.2	(20)
(21)		2%	DB	46.5	50.5	56.8	65.5	74.5	79.7	83.2	83.6	75.6	65.7	52.9	49.7	(21)
(22)			MCWB	42.9	45.2	47.9	52.0	59.1	62.8	65.3	66.0	62.0	56.0	48.5	45.3	(22)
(23)		5%	DB	44.2	45.5	51.5	61.0	70.6	75.5	79.3	79.1	71.7	61.7	49.9	46.7	(23)
(24)			MCWB	41.0	41.1	44.7	49.8	57.1	61.9	64.0	63.6	60.2	54.0	46.1	43.5	(24)
(25)		10%	DB	42.0	41.1	48.4	57.0	67.1	71.3	76.0	75.3	67.7	58.4	47.3	44.0	(25)
(26)			MCWB	39.0	37.8	43.2	48.0	55.2	59.5	62.3	62.2	58.5	52.5	44.1	41.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.6	49.2	52.2	55.9	63.3	67.7	69.5	69.2	65.8	60.9	51.7	48.5	(27)
(28)			MCD B	49.3	55.2	60.8	68.2	74.5	78.6	83.8	85.1	75.5	71.0	55.5	51.5	(28)
(29)		2%	WB	43.5	45.4	48.2	53.0	60.2	64.7	66.9	66.6	63.2	57.5	49.0	45.8	(29)
(30)			MCD B	46.3	49.8	55.9	62.9	71.3	75.5	79.5	80.6	72.6	64.3	52.6	49.2	(30)
(31)		5%	WB	41.3	41.2	45.7	50.8	58.2	62.6	64.9	64.6	61.3	54.9	46.2	43.5	(31)
(32)			MCD B	43.8	45.1	50.8	59.5	68.7	72.0	76.9	76.9	70.0	60.2	49.4	46.5	(32)
(33)		10%	WB	39.0	38.2	43.4	48.4	56.0	60.6	63.2	62.8	59.3	52.9	44.2	41.4	(33)
(34)			MCD B	41.8	41.0	48.0	56.0	65.8	70.8	74.3	73.0	66.6	57.7	47.1	44.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.0	11.7	13.0	16.1	19.1	17.4	19.2	18.9	17.3	14.3	10.1	7.9	(35)
(36)			MCDBR	10.1	16.2	19.8	24.4	27.4	27.1	27.3	27.0	24.6	19.9	14.0	9.6	(36)
(37)			MCWBR	8.6	11.6	13.2	14.3	15.0	13.5	13.4	12.9	13.4	12.2	10.6	8.3	(37)
(38)		5% WB	MCDBR	9.4	15.2	19.1	21.5	24.3	23.8	24.2	23.6	21.6	17.5	12.8	9.1	(38)
(39)			MCWBR	8.5	11.3	13.3	13.2	14.2	12.5	13.0	11.9	12.3	11.5	10.4	8.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.298	0.321	0.363	0.401	0.404	0.404	0.417	0.410	0.377	0.351	0.324	0.294	(40)	
(41)		taud	2.386	2.343	2.296	2.242	2.268	2.292	2.255	2.296	2.368	2.427	2.441	2.403	(41)	
(42)		Ebn at Noon	234	257	265	266	271	271	265	261	257	243	222	217	(42)	
(43)		Edh at Noon	20	27	34	40	41	40	41	37	31	25	19	17	(43)	
(44)	All-Sky Solar Radiation	RadAvg	253	498	815	1282	1531	1665	1583	1374	983	581	276	194	(44)	
(45)		RadStd	47	75	114	167	184	143	178	136	139	74	36	23	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (50 neighbors)	+0.72	N/A	N/A	N/A	+0.58	+0.67	N/A	N/A	+108	+36	

Nomenclature: See separate page